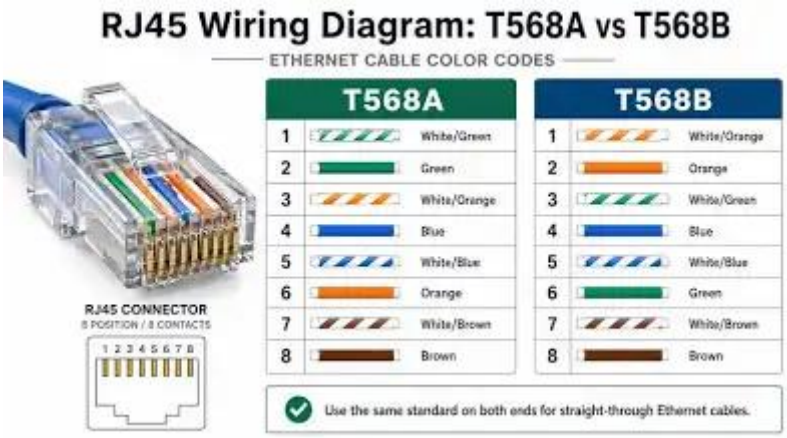
Network cable and connectors

Q-1 Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.

Ans:



Q-2 List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.

Ans: **Three Types of Network Cables and Their Uses**

Network Cable	**Real-World App or Device**	**Why It Is Used**
1. Cat5e Ethernet Cable	Wi-Fi Router	It connects a router to a computer or other network device and supports speeds up to 1 Gbps over suitable distances.
2. Cat6 Ethernet Cable	Zomato POS Terminal	It connects a restaurant's billing terminal to the local network for reliable communication and order processing.
3. Fiber Optic Cable	IPL Streaming Server	It transmits data using light at very high speeds over long distances, making it suitable for carrying large amounts of video data.

Q-3 Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.

Ans: **RJ-45 Connector Crimping – Step-by-Step Process**

YouTube video: How to Crimp RJ45 Connector

YouTube

Tools Required

- Ethernet cable (Cat5e or Cat6)
- RJ-45 connector
- Crimping tool
- Cable stripper or cutter
- LAN cable tester

Step-by-Step Process

1. Cut the cable: Cut the Ethernet cable to the required length using a cable cutter.

2. Remove the outer covering: Strip about 2–3 cm of the outer jacket carefully without damaging the inner wires.

3. Separate the wires: Untwist the four pairs and straighten the eight small wires.

4. Arrange the wires: Arrange the wires in the T568B colour order:

White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown.



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5. Trim the wires: Cut all eight wires evenly so they can fit properly inside the connector.

6. Insert the wires: Push all eight wires into the RJ-45 connector. Make sure each wire reaches the front end and the outer cable jacket enters the connector.

7. Crimp the connector: Place the connector into the crimping tool and press the handles firmly to secure the wires.

8. Repeat and test: Crimp the other end using the same colour order for a standard straight-through cable. Use a LAN cable tester to check that all eight wires are connected correctly.

Two Common Mistakes to Avoid

1. Incorrect wire order: Arranging the wires in the wrong colour sequence can cause the cable to fail. Always check the T568B order before crimping.

2. Wires not inserted fully: If the wires do not reach the end of the connector, the connection may not work properly. Check all eight wires before crimping.



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Q-4 Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.

Ans: Cable and Connector Selection for an IPL Gaming Zone

For an IPL screening event, I would choose the following cables and connectors to connect multiple screens, gaming consoles, and network devices with high speed and reliable performance.

Cable Type	**Connector Type**	**Usage**	**Reason**
HDMI 2.1 Cable	HDMI Connector	Connect gaming consoles to TVs or monitors	Supports high-quality video and audio with low delay.
Cat6 Ethernet Cable	RJ-45 Connector	Connect consoles, PCs, and network switches to the router	Provides a stable, high-speed wired network connection with less interference than Wi-Fi.
Fiber Optic Cable	Fiber Optic Connector	Connect the main network to a distant screening area or network switch	Supports high-speed data transfer over long distances with low signal loss.

Explanation of Choices

1. HDMI 2.1: Used to transfer video and audio from gaming consoles to screens. It supports high resolutions and refresh rates when the connected devices are compatible.

2. Cat6 Ethernet: Used for wired internet connections. It helps reduce lag and provides stable connectivity during online multiplayer gaming and live IPL streaming.

3. Fiber Optic: Used for connecting network equipment over longer distances, especially when high bandwidth is required for multiple screens and devices.

Q-5 Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WiFi router or at a café's POS counter), and describe what type it is and what devices it connects.

Hint: Look for blue or yellow cables with clear plastic ends (RJ-45) or thin round cables for fiber optics.

Ans: **Photo of a Network Cable in Use**

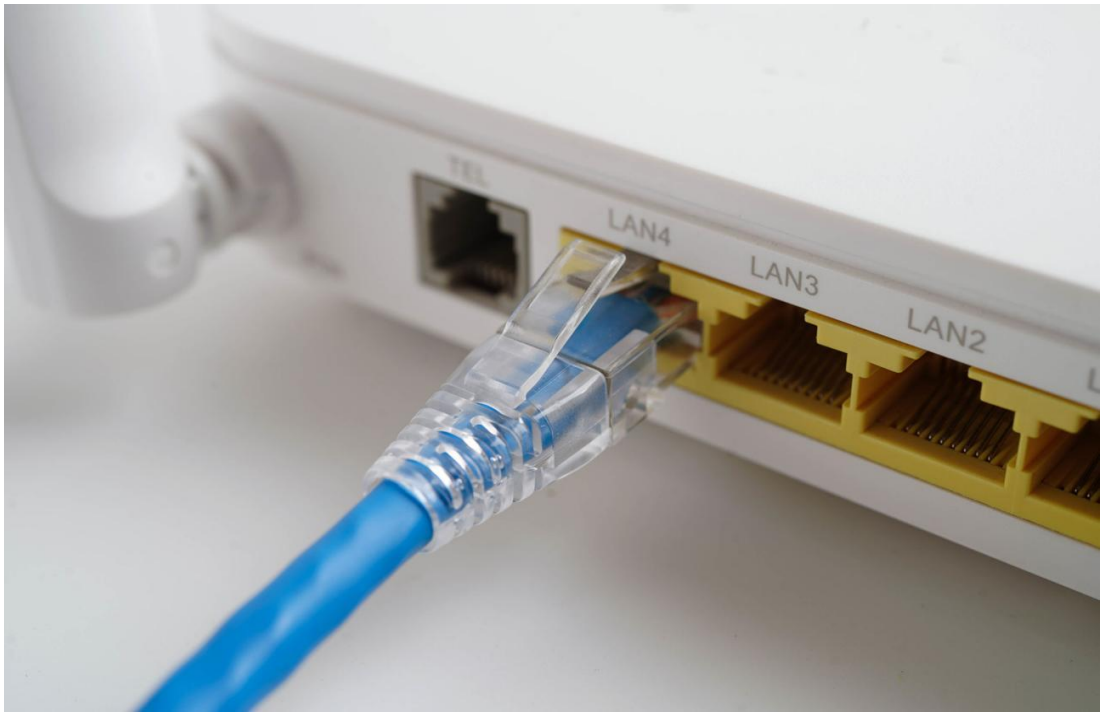


Photo source: View the original photo on Unsplash.

Free Communication Image on Unsplash

Description

Type of cable: Ethernet LAN Cable (Cat5e or Cat6)

Connector type: RJ-45

Devices connected: Wi-Fi router and a computer, switch, or another network device.

Purpose: The Ethernet cable provides a wired network connection. It helps transfer data reliably and provides stable internet access for activities such as online classes, video streaming, and gaming.

Conclusion: An Ethernet cable with an RJ-45 connector is commonly used to connect network devices and provide a reliable wired internet connection.

Note: If your assignment requires a photo from your own home or college, take a picture of a LAN cable plugged into a router and attach it instead.